数学花园
探秘
MATHEMATICS GARDEN

2022 Spring Cup
Mathematical Olympiad

PRELIMINARY ROUND

Date: 28 January 2023

Time Given: 1 hour 30 minutes

Level: Secondary Junior

Name: _____

Instructions to Candidates

1. Do not open the booklet until you are told to do so.
2. Answer ALL 13 questions.
3. Write your answers in the answer sheet provided.
4. No steps are needed to justify your answers.
5. Questions 1-4 are worth 8 marks each.
6. Questions 5-8 are worth 10 marks each.
7. Questions 9-11 are worth 12 marks each.
8. Questions 12-13 are worth 15 marks each.
9. No marks will be deducted for wrong answers.
10. No marks will be given for unanswered questions.

Part 1 Short Answer Questions I (8 Marks Each, Total 32 Marks)

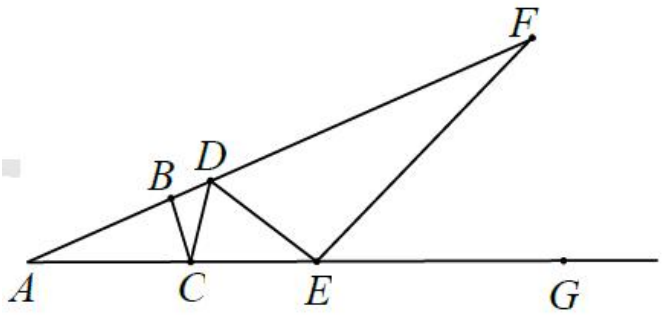
1. Simplify $\dfrac{\dfrac{4}{3^2}\times\dfrac{7}{4^2}}{\left(\dfrac{1}{3^2}+\dfrac{7}{8^2}\right)^2-\left(\dfrac{1}{3^2}-\dfrac{7}{8^2}\right)^2}+\left|2-\pi\right|^0-(-2)^3$.

Answer: _____

2. Given that the algebraic expression ax^3+bx+c has a value of 2 when $x=0$ and a value of -145 when $x=-3$, find the value of the expression when $x=3$.

Answer: _____

3. As shown in the figure, given that $\angle A=18°$, points B, C, D, E, F, and G all lie on the two sides of $\angle A$, $\angle ABC=89°$, $\angle ACB=\angle DCE$, $\angle ADC=\angle EDF$, $\angle CED=\angle FEG$, find the value of $\angle F$.



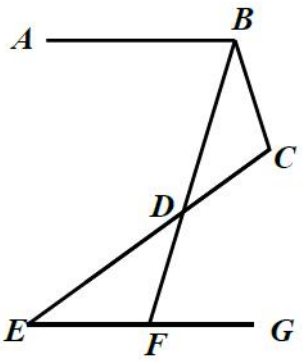
Answer: _____

4. Given that $a^2+3a+1=0$, and $a^4+ma^2+1=11(a^3+ma^2+a)$, find the value of the real number m.

Answer: _____

Part 2 Short Answer Questions II (10 Marks Each, Total 40 Marks)

5. As shown in the figure, $AB//EG$, $\angle ACB=\angle C=\angle BFE$, $\angle CBF=\angle EDF$. Find the value of $\angle C$.



Answer: _____

6. Given that two systems of equations in x and y $\begin{cases} ax-3y=2 \\ 4x+y=32 \end{cases}$ and $\begin{cases} 3x+by=29 \\ 7x-9y=13 \end{cases}$ have the same solution, find the value of $a+b$.

Answer: _____

7. Given that $a_1, a_2, ..., a_9$ is an arrangement of 1, 2, …, 9,

   $S = |a_1 - 2a_2| + |3a_3 - 4a_4| + |5a_5 - 6a_6| + + |7a_7 - 8a_8| + |9a_9 - 10|$, find the minimum value of *S*.

Answer: _____

8. Two prime numbers *p* and *q* form the expression $p^2 - 4q^4$ which is a perfect square. How many pairs of (*p, q*) are there?

Answer: _____

Part 3 Short Answer Questions III (12 Marks Each, Total 36 Marks)

9. Line *a* is parallel to line *b*, with 20 points on line *a* and 22 points on line *b*. Connect these 42 points as vertices to form 14 triangles such that each point is used exactly once, and no two triangles share any common sides. How many different ways are there to connect them?

Answer: _____

10. Writing a two-digit number before a three-digit number results in a five-digit number, which is exactly the square of the sum of the original two numbers. What is this five-digit number?

Answer: _____

11. A and B take turns writing $2n$ natural numbers on the blackboard. Each time, they may only write one of the numbers 1, 2, 3, or 4. A goes first, and each player writes n numbers. If the sum of the $2n$ natural numbers at the end is a multiple of 6, then B wins; otherwise, A wins. Assuming both players are sufficiently intelligent, among the positive integers n not exceeding 2022, for how many values of n does A have a guaranteed winning strategy?

Answer: _____

Part 4 Full Length Questions (15 Marks Each, Total 30 Marks)

12. There are real numbers *a*, *b*, and *c* where $a^2+b^2+c^2=100$, $(a-b)^2 \geq 50$, $(b-c)^2 \geq 50$, and $(c-a)^2 \geq 50$, find the value of $a^4+b^4+c^4$.

Answer: _____

13. On a sheet of paper there are *n* distinct rational numbers. It is known that for any three different numbers chosen from them, at least two of them have a product that is also one of the numbers on the paper. Find the maximum possible value of n.

Answer: _____